

MH1812 Discrete Mathematics

- Continuous Assessment 1

NTU, AY17/18 S1

21 Sept 2017, 10:30AM - 11:20AM

Name:

Tutorial Group:

NTU Email:

*Note: **first 3** questions are compulsory. Question 4 is bonus, you get the points if you answer it correctly, you don't lose any points if you leave it or answer wrongly. You can write on the back of paper if there is not enough space, leave the tutorial group if you can't remember.*

Question 1 (30 points)

1. Compute 20^{2017} modulo 4 (10 points).

Solution: Since $20 \equiv 0 \pmod{4}$, we have $20^{2017} \equiv 0^{2017} \equiv 0 \pmod{4}$. $\square$

2. Consider the set S of multiples of 4 that is $S = \{\dots, -12, -8, -4, 0, 4, 8, 12, \dots\}$. Is the set S closed under multiplication? Justify your answer (10 points).

Solution: Yes. For any elements $4a$ and $4b$ in S , we have $4a \times 4b = 4(4ab)$, which is also an element in S . $\square$

3. Let $S = \{2, 3, 4, 5\}$, and operator Δ be "addition modulo 6", i.e., $a\Delta b = (a+b) \pmod{6}$. Is S closed under Δ ? Justify your answer. (10 points)

Solution: No. There exist $a, b \in S$ such that $a+b \pmod{6} \notin S$. For example, $3\Delta 4 = 7 \pmod{6} = 1 \pmod{6}$, but 1 is not in S . $\square$

Question 2 (40 points)

1. Prove or disprove the following statement (20 points):

$$(p \rightarrow (q \rightarrow r)) \equiv ((p \wedge q) \rightarrow r).$$

Solution: Yes, The statement is true.

$$\begin{aligned} (p \rightarrow (q \rightarrow r)) &\equiv (\neg p \vee (q \rightarrow r)) \quad (\text{Conversion}) \\ &\equiv \neg p \vee (\neg q \vee r) \quad (\text{Conversion}) \\ &\equiv (\neg p \vee \neg q) \vee r \quad (\text{Associativity}) \\ &\equiv \neg(p \wedge q) \vee r \quad (\text{De Morgan}) \\ &\equiv (p \wedge q) \rightarrow r. \quad (\text{Conversion}) \end{aligned}$$

Another solution is to show that the truth values of the two statements are the same by the truth table. □

2. Decide whether the following argument is valid (20 points):

$$\begin{array}{l} p \\ p \rightarrow q \\ \neg q \vee r \\ \therefore r \end{array}$$

Solution: Since p and $p \rightarrow q$ are true. It follows q is true. Therefore, $\neg q$ is false. But $\neg q \vee r$ is true and $\neg q$ is false. r must be true. So the argument valid.

Alternatively, we can show it by the truth table.

p	q	r	$\neg q$	p	$p \rightarrow q$	$\neg q \vee r$	r
T	T	T	F	T	T	T	T
T	T	F	F	T	T	F	
T	F	T	T	T	F	T	
T	F	F	T	T	F	T	
F	T	T	F	F	T	T	
F	T	F	F	F	T	F	
F	F	T	T	F	T	T	
F	F	F	T	F	T	T	

The only critical row is the first row in which all the premises are true. The conclusion is also true, so the argument is valid. □

Question 3 (30 points)

Given sets $A = \{3, 4\}$, $B = \{2, 3, 5\}$, and $P(x, y)$ denotes " $x^2 - y^2 \geq 5$ ", determine the truth value of the following statement and justify your answer :

1. $\forall x \in A, \exists y \in B, P(x, y)$ (15 points).

Solution: The statement is true. Because, when $x = 3$, take $y = 2$, then $3^2 - 2^2 = 5 \geq 5$; when $x = 4$, take $y = 3$, then $4^2 - 3^2 = 7 \geq 5$, we show that the statement is true. $\square$

2. $\exists x \in A, \forall y \in B, P(x, y)$ (15 points).

Solution: The statement is false. If $x = 3$, take $y = 3$, then $3^2 - 3^2 = 0 < 5$. If $x = 4$, take $y = 5$, then $4^2 - 5^2 = -9 < 5$. Therefore, no such x exists such that $\forall y \in B, P(x, y)$ is true. $\square$

Question 4 (Bonus Question) (10 points)

Prove that there are infinitely many prime numbers.

Solution: We prove it by contradiction. Suppose there are only finitely many prime numbers, say, $p_1, p_2, \dots, p_k$. Let $N = p_1 p_2 \dots p_k + 1$. Then N is not one of any k prime numbers. It follows that N is not a prime number, so there exists a prime number, say, p_i such that $p_i | N$. We then assume that $N = p_i a$ for some a . We then have $N = p_i a = p_i(p_1 \dots p_{i-1} p_{i+1} \dots p_k) + 1$. We then have $1 = p_i(a - p_1 \dots p_{i-1} p_{i+1} \dots p_k)$, which is not possible. A contradiction! $\square$